Practical No. 9: Fitting of Regression Lines and other curves

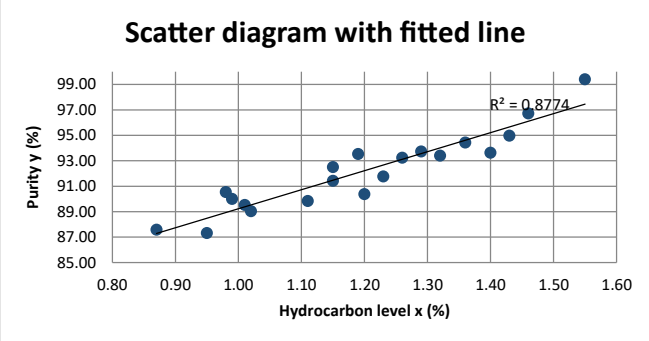
Blue = input data from question; black = formulas. Least-squares method via normal equations (Σ sums shown in Total rows).

Ex.1 Oxygen purity (y) vs hydrocarbon level (x): scatter diagram and simple linear regression y = b0 + b1·x

Obs. No.	x (%)	y (%)	x·y	x²
1	0.99	90.01	89.1099	0.9801
2	1.02	89.05	90.8310	1.0404
3	1.15	91.43	105.1445	1.3225
4	1.29	93.74	120.9246	1.6641
5	1.46	96.73	141.2258	2.1316
6	1.36	94.45	128.4520	1.8496
7	0.87	87.59	76.2033	0.7569
8	1.23	91.77	112.8771	1.5129
9	1.55	99.42	154.1010	2.4025
10	1.40	93.65	131.1100	1.9600
11	1.19	93.54	111.3126	1.4161
12	1.15	92.52	106.3980	1.3225
13	0.98	90.56	88.7488	0.9604
14	1.01	89.54	90.4354	1.0201
15	1.11	89.85	99.7335	1.2321
16	1.20	90.39	108.4680	1.4400
17	1.26	93.25	117.4950	1.5876
18	1.32	93.41	123.3012	1.7424
19	1.43	94.98	135.8214	2.0449
20	0.95	87.33	82.9635	0.9025
Total	**23.9200**	**1843.2100**	**2214.6566**	**29.2892**

Expected: y = 74.283 + 14.947x

Result	Value
n	20
b1 = (nΣxy−ΣxΣy)/(nΣx²−(Σx)²)	**14.947**
b0 = ȳ − b1·x̄	**74.283**
Fitted model	**y = 74.283 + 14.947x**
Check: SLOPE	14.947
Check: INTERCEPT	74.283



Ex.2 Straight line y = a + bx: sales revenue increase (y) vs advertising increase (x); estimate at x = 7.5

Firm	x (%)	y (%)	x·y	x²
A	1	1	1	1
B	3	2	6	9
C	4	2	8	16
D	6	4	24	36
E	8	6	48	64
F	9	8	72	81
G	11	8	88	121
H	14	9	126	196
Total	**56**	**40**	**373**	**524**

Expected: increase in sales revenue ≈ 5.352%

Result	Value
n	8
b = (nΣxy−ΣxΣy)/(nΣx²−(Σx)²)	**0.7045**
a = ȳ − b·x̄	**0.0682**
Fitted line	**y = 0.0682 + 0.7045x**
x for estimate (input)	7.5
Estimated y (% increase)	**5.352**

Ex.3 Second-degree polynomial y = a + bx + cx² (normal equations solved for a, b, c)

x	x²	y	x³	x⁴	x·y	x²·y
1	1	6	1	1	6	6
2	4	11	8	16	22	44
3	9	18	27	81	54	162
4	16	27	64	256	108	432
10	**30**	**62**	**100**	**354**	**190**	**644**

Normal equations (coefficients of a, b, c | RHS):

a	b	c	RHS	Check
4	10	30	62	0.000
10	30	100	190	0.000
30	100	354	644	0.000

Result	Value
c (coeff. of x²)	**1.0000**
b (coeff. of x)	**2.0000**
a (constant)	**3.0000**
Fitted curve	**y = 3.00 + 2.00x + 1.00x²**

Check column = LHS − RHS using the solved a, b, c (should be 0). Expected: y = 3 + 2x + x²

Ex.4 Exponential curve y = a·b^x (linearised: Y = ln y = ln a + x·ln b)

x	y	Y = ln y	x·Y	x²
1	1.0	0.0000	0.0000	1
2	1.2	0.1823	0.3646	4
3	1.8	0.5878	1.7634	9
4	2.5	0.9163	3.6652	16
5	3.6	1.2809	6.4047	25
6	4.7	1.5476	9.2854	36
7	6.6	1.8871	13.2095	49
8	9.1	2.2083	17.6662	64
36		**8.6102**	**52.3589**	**204**

Expected: y = 0.6821(1.38)^x

Result	Value
n	8
B = ln b = (nΣxY−ΣxΣY)/(nΣx²−(Σx)²)	0.3241
A = ln a = Ȳ − B·x̄	-0.3822
a = e^A	**0.6823**
b = e^B	**1.3828**
Fitted curve	**y = 0.6823(1.38)^x**